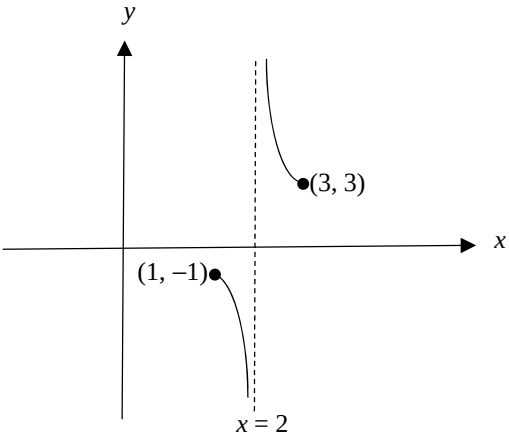
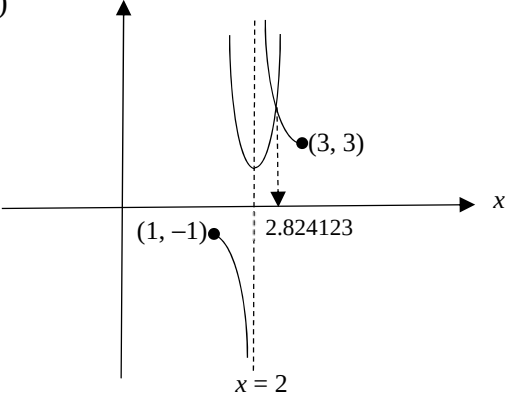
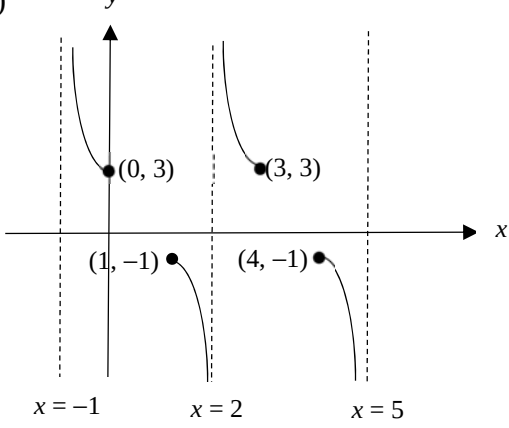


Solutions to 2018 Y6 H2 Maths Prelim Exam P2

1.	Solution [10 marks]
	(i) $y = \frac{x^2 - 3x + 3}{x - 2} = x - 1 + \frac{1}{x - 2}$ $\frac{dy}{dx} = 1 - \frac{1}{(x - 2)^2}$ For $x = 1$ or 3, $\frac{dy}{dx} = 0$ $\Rightarrow C$ has stationary points at $x = 1$ and $x = 3$. For $1 < x < 2$ or $2 < x < 3$, $\frac{dy}{dx} < 0$ $\Rightarrow C$ is decreasing on $1 < x < 2$ and $2 < x < 3$. <u>Alternatively,</u> Since $1 \leq x \leq 3$, $0 \leq (x - 2)^2 \leq 1$ $\therefore \frac{1}{(x - 2)^2} \geq 1$ $\therefore \frac{dy}{dx} \leq 0$ i.e. the gradient is always non-positive.
	(ii) 

	(iii)  $\frac{x^2 - 4x + 5}{x - 2} \geq 3(x - 2)^2$ $\frac{x^2 - 3x + 3 - x + 2}{x - 2} \geq 3(x - 2)^2$ $\frac{x^2 - 3x + 3}{x - 2} \geq 3(x - 2)^2 + 1$ Sketch the addition graph $y = 3(x - 2)^2 + 1$. Thus, $2 < x \leq 2.82$.	
	(iv) 	

2.	Solution [10 marks]	
	(i) Let $\frac{2(2r+3)}{(r+1)(r+2)(r+3)} = \frac{A}{r+1} + \frac{B}{r+2} + \frac{C}{r+3}$. By cover up rule, $A = \frac{2(1)}{(1)(2)} = 1; B = \frac{2(-1)}{(-1)(1)} = 2; C = \frac{2(-3)}{(-2)(-1)} = -3;$ $\therefore \frac{2(2r+3)}{(r+1)(r+2)(r+3)} = \frac{1}{r+1} + \frac{2}{r+2} - \frac{3}{r+3}$	
	(ii) $\sum_{r=1}^n \frac{2r+3}{(r+1)(r+2)(r+3)} = \frac{1}{2} \sum_{r=1}^n \frac{2(2r+3)}{(r+1)(r+2)(r+3)}$ $= \frac{1}{2} \sum_{r=1}^n \left(\frac{1}{r+1} + \frac{2}{r+2} - \frac{3}{r+3} \right)$ $= \frac{1}{2} \left[\begin{array}{ccc} \frac{1}{2} & + & \frac{2}{3} & - & \frac{3}{4} \\ \frac{1}{3} & + & \frac{2}{4} & - & \frac{3}{5} \\ \frac{1}{4} & + & \frac{2}{5} & - & \frac{3}{6} \\ \dots & \dots & \dots & \dots & \dots \\ \frac{1}{n-1} & + & \frac{2}{n} & - & \frac{3}{n+1} \\ \frac{1}{n} & + & \frac{2}{n+1} & - & \frac{3}{n+2} \\ \frac{1}{n+1} & + & \frac{2}{n+2} & - & \frac{3}{n+3} \end{array} \right]$ $= \frac{1}{2} \left[\frac{3}{2} - \frac{1}{n+2} - \frac{3}{n+3} \right]$ $= \frac{3}{4} - \frac{1}{2(n+2)} - \frac{3}{2(n+3)}$	

	(iii) $\frac{21}{3 \times 4 \times 5} + \frac{27}{4 \times 5 \times 6} + \frac{33}{5 \times 6 \times 7} + \dots$ $= 3 \left(\frac{7}{3 \times 4 \times 5} + \frac{9}{4 \times 5 \times 6} + \frac{11}{5 \times 6 \times 7} + \dots \right)$ $= 3 \sum_{r=2}^{\infty} \frac{2r+3}{(r+1)(r+2)(r+3)}$ $= 3 \left[\sum_{r=1}^{\infty} \frac{2r+3}{(r+1)(r+2)(r+3)} - \frac{2(1)+3}{(1+1)(1+2)(1+3)} \right]$ $= 3 \lim_{n \rightarrow \infty} \left[\frac{3}{4} - \frac{1}{2(n+2)} - \frac{3}{2(n+3)} \right] - \frac{5}{8}$ As $n \rightarrow \infty$, $\frac{3}{2(n+2)} \rightarrow 0$ and $\frac{9}{2(n+3)} \rightarrow 0$. Hence the series is convergent, and the sum = $3 \left[\frac{3}{4} - 0 - 0 \right] - \frac{5}{8} = \frac{13}{8}$	
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3.	Solution [10 marks]	
	(i) $\cos^{-1} \left(\frac{\begin{pmatrix} -1 \\ 0 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}}{\sqrt{17}\sqrt{5}} \right) = \cos^{-1} \frac{7}{\sqrt{17}\sqrt{5}} = 40.6013^\circ$ Angle between roofs $= 180^\circ - 40.6013^\circ = 139.4^\circ$	

	(ii) Plane $ABGF$: $\mathbf{r} \cdot \begin{pmatrix} -1 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 20 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 0 \\ 4 \end{pmatrix} = 80$ Plane $BCHG$: $\mathbf{r} \cdot \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 50 \\ 0 \\ 25 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = 100$ Taking intersection and using GC, $l_{BG} : \mathbf{r} = \begin{pmatrix} 40 \\ 0 \\ 30 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \lambda \in \mathbb{R}$ When $y = 50, \lambda = 50$. Thus coordinates of sensor is $(40, 50, 30)$.	
	(iii) Coordinates of B is $(40, 0, 30)$. Cable is anchored to the ground at the point $P(25, -10, 0)$. Required length = perpendicular distance from P to line AB. $\vec{PA} = \begin{pmatrix} 0 \\ 0 \\ 20 \end{pmatrix} - \begin{pmatrix} 25 \\ -10 \\ 0 \end{pmatrix} = \begin{pmatrix} -25 \\ 10 \\ 20 \end{pmatrix}$ $\vec{AB} = \begin{pmatrix} 40 \\ 0 \\ 30 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 20 \end{pmatrix} = \begin{pmatrix} 40 \\ 0 \\ 10 \end{pmatrix}$ Length of shortest cable $= \frac{\left \vec{PA} \times \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} \right }{\left \begin{pmatrix} -25 \\ 10 \\ 20 \end{pmatrix} \times \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} \right }$ $= \frac{\left \begin{pmatrix} 10 \\ 105 \\ -40 \end{pmatrix} \right }{\sqrt{17}} = 27.4 \text{ metres}$	

	(iv) The thickness of materials used for building is negligible.	
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4.	Solution [10 marks]	
	(i) f^{-1} exists because f is one-one on $x < m < 2$. $D_f = (-\infty, m), R_f = \left(0, \frac{1}{(m-2)^2}\right)$ For f^2 to exist, $R_f \subseteq D_f$. $0 < \frac{1}{(m-2)^2} \leq m$ $m(m-2)^2 \geq 1$ $m^3 - 4m^2 + 4m - 1 \geq 0$ Solving with GC, $0.382 \leq m \leq 1$ or $m \geq 2.84$ (reject since $m < 2$)	
	(ii) $y = \frac{1}{(x-2)^2} \Rightarrow x = 2 \pm \frac{1}{\sqrt{y}}$ Since $x < m < 2$, $x = 2 - \frac{1}{\sqrt{y}}$. $f^{-1}(x) = 2 - \frac{1}{\sqrt{x}}$ $D_{f^{-1}} = \left(0, \frac{1}{(m-2)^2}\right)$ $R_{f^{-1}} = (-\infty, m)$ $f^2(x)$ $= f\left(\frac{1}{(x-2)^2}\right)$ $= \frac{1}{\left(\frac{1}{(x-2)^2} - 2\right)^2}$ $= \frac{(x-2)^4}{(1-2(x-2)^2)^2}$	

	$D_{f^2} = (-\infty, m)$ $(-\infty, m) \xrightarrow{f} \left(0, \frac{1}{(m-2)^2} \right) \xrightarrow{f} \left(\frac{1}{4}, \frac{(m-2)^4}{(1-2(m-2)^2)^2} \right) = R_{f^2}$	
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5.

Solution [7 marks]

(i) Using the following possibility table, we have:

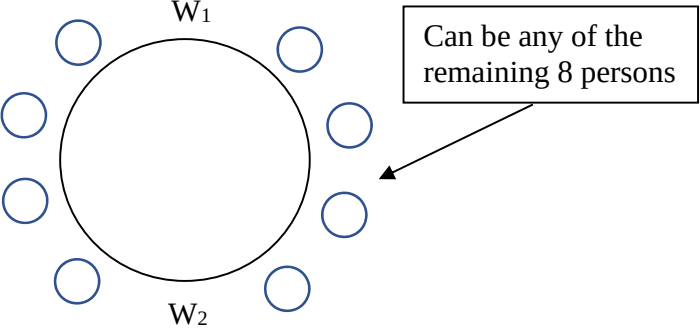
		2 nd die					
		1	2	3	4	5	6
1 st die	1	1	1	1	1	1	1
	2	1	2	2	2	2	2
	3	1	2	3	3	3	3
	4	1	2	3	4	4	4
	5	1	2	3	4	5	5
	6	1	2	3	4	5	6

From the table, $P(X = 2) = \frac{9}{36} = \frac{1}{4}$

Probability distribution of X :

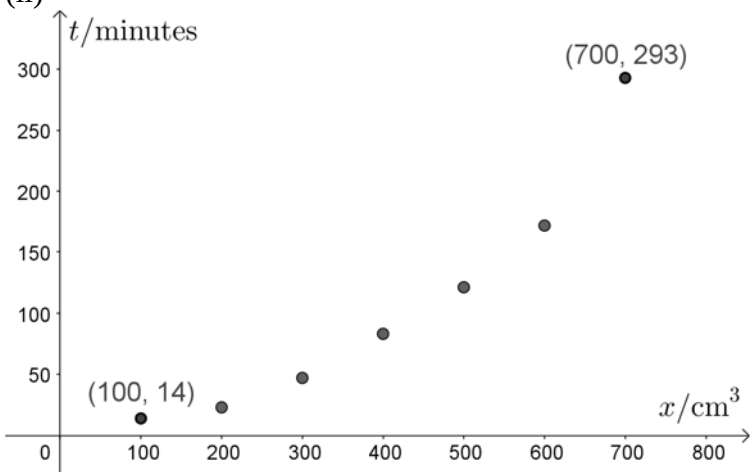
x	1	2	3	4	5	6
$P(X = x)$	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{5}{36}$	$\frac{3}{36}$	$\frac{1}{36}$

(ii) $E(X) = \sum_{x=1}^6 xP(X=x) = \frac{91}{36} \quad (\text{or } 2.53)$ $\text{Var}(X) = E(X^2) - (E(X))^2$ $= \left(\sum_{x=1}^6 x^2 P(X=x) \right) - \left(\sum_{x=1}^6 x P(X=x) \right)^2$ $= \frac{301}{36} - \left(\frac{91}{36} \right)^2$ $= \frac{2555}{1296} \quad (\text{or } 1.97)$	
(iii) $P(\text{Ivan wins}) = \left(\frac{1}{4} \right) + \left(\frac{3}{4} \right)^2 \left(\frac{1}{4} \right) + \left(\frac{3}{4} \right)^4 \left(\frac{1}{4} \right) + \dots$ $= \frac{\frac{1}{4}}{1 - \left(\frac{3}{4} \right)^2} = \frac{4}{7}$ $E(\text{gain}) \text{ for Ivan} = \frac{4}{7} \times 7 + \frac{3}{7} \times (-10) = -\0.29 Since Ivan is expecting to lose 29 cents, Jon has a higher expected gain. <u>Alternatively,</u> $P(\text{Jon wins}) = \left(\frac{3}{4} \right) \left(\frac{1}{4} \right) + \left(\frac{3}{4} \right)^3 \left(\frac{1}{4} \right) + \left(\frac{3}{4} \right)^5 \left(\frac{1}{4} \right) + \dots$ $= \frac{\left(\frac{3}{4} \right) \left(\frac{1}{4} \right)}{1 - \left(\frac{3}{4} \right)^2} = \frac{3}{7}$ $E(\text{gain}) \text{ for Jon} = \frac{3}{7} \times 10 + \frac{4}{7} \times (-7) = \0.29 Since Jon is expecting to win 29 cents (i.e. Ivan is expected to lose 29 cents), he has a higher expected gain.	

6.	Solution [8 marks]	
	(a) (i) Number of ways = $10! = 3\,628\,800$ ways (ii) $\uparrow W_1 \uparrow W_2 \uparrow W_3 \uparrow W_4 \uparrow W_5 \uparrow W_6 \uparrow$ $\uparrow$ Position for each of the men Number of ways = $10! - \text{No. of ways all men are separated}$ $= 10! - 6 \times \binom{7}{4} 4!$ $= 3\,628\,800 - 604\,800$ $= 3\,024\,000$ ways (iii) $\uparrow M_1 \uparrow M_2 \uparrow M_3 \uparrow M_4 \uparrow$ $\uparrow$ Position for the following “items” $\boxed{W_1 W_2 W_3 W_4 W_5}, W_6$ the w group Number of ways = $4! \times \binom{5}{2} 2! \times \binom{6}{5} 5!$ $= 345\,600$ ways	
	(b) Let particular women be W_1, W_2  Can be any of the remaining 8 persons Fix W_1 and W_2, then arrange the remaining 8 persons. $\Rightarrow$ No. of ways = $8!$ $\therefore$ Required probability $= \frac{8!}{(10-1)!} = \frac{1}{9}$	

7.	Solution [10 marks]	
	(i) Let X be the r.v. “no of customers who uses cashless payment” out of 30. $X \sim B\left(30, \frac{p}{100}\right)$ $P(X \leq 1) = 0.245$ $P(X = 0) + P(X = 1) = 0.245$ $\binom{30}{0} \left(1 - \frac{p}{100}\right)^{30} + \binom{30}{1} \left(\frac{p}{100}\right) \left(1 - \frac{p}{100}\right)^{29} = 0.245$ $\left(1 - \frac{p}{100}\right)^{30} + 0.3p \left(1 - \frac{p}{100}\right)^{29} - 0.245 = 0$ Using Solver in GC, $p = 8.811$	
	(ii) Let Y be the r.v. “no of customers who use cashless payment” out of 29. $Y \sim B(29, 0.3)$ Required Probability $= P(Y = 9) \times 0.3$ $= 0.0471872337$ ≈ 0.0472	
	(iii) $X \sim B(30, 0.3)$ Let W be the r.v. “no of customers who use cashless payment” out of 60. $W \sim B(60, 0.3)$ Requires probability $= \frac{P(X_1 = 13, X_2 \leq 2) + P(X_1 = 14, X_2 \leq 1) + P(X_1 = 15, X_2 = 0)}{P(W < 16)}$ $= \frac{2 \times (P(X = 13)P(X \leq 2) + P(X = 14)P(X \leq 1) + P(X = 15)P(X = 0))}{P(W \leq 15)}$ $= 0.0008312264247$ ≈ 0.000831	

	(iv) The assumption is that the people being surveyed are all different in both samples. (OR Samples are independently and identically distributed.)	
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8.	Solution [10 marks]	
	(i) $\bar{x} = 400$ $\bar{t} = -65.1429 + 0.4318(400)$ $\bar{t} = 107.5771$ $14 + 23 + 47 + 83 + a + 172 + 293 = 7 \times \bar{t}$ $632 + a = 753.0397$ $a \approx 121 \text{ (shown)}$	
	(ii)  $r = 0.9414677299 \approx 0.941$ (3 sig fig)	
	(iii) (a) Although the r value is 0.941 which is close to 1 suggesting that there is a strong linear correlation, the scatter diagram shows that the relationship is more of a curvilinear one, with t increasing increasingly as x increases. Thus, the regression line is not appropriate. (b) The regression line predicts that for volume less than a certain amount, it takes negative time for the water to cool to room temperature which is not possible in the context. Thus, the regression line is not appropriate.	

	(iv) Calculating the r- values for models A and B, Model A: r-value = 0.987 while Model B: $t = ae^{bx} \Rightarrow \ln t = \ln a + bx$, r-value = 0.994, Since the r-value for model B is closer to 1 compared to model A, model B is the better model. Regression line of $\ln t$ on x is $\ln t = 2.224858298 + 0.0050332105x$ $\ln t = 2.22 + 0.00503x$ (3 sig fig)	
	(v) When $x = 450$, $\ln t = 2.224858298 + 0.0050332105(450)$ $t = 89.10389273$ ≈ 89.1 minutes (3 sig fig) The estimate is reliable as the appropriate regression line is used (t being the dependent variable); $r = 0.994$ which is close to 1, suggesting a strong linear correlation between $\ln t$ and x; and the estimated point where $x = 450$ is within the data range of 100 to 700.	

9.	Solution [12 marks]	
	(i) $\bar{x} = \frac{207000}{60} = 3450$ $\Sigma(x - 3400) = 3000$ $s^2 = \frac{1}{59} \left[5450000 - \frac{3000^2}{60} \right] = 89830.50847 \approx 89800$	

	(ii) Let μ be the mean starting salaries of fresh graduates. Test $H_0: \mu = 3500$ (i.e. manager's belief is incorrect) Against $H_1: \mu \neq 3500$ (i.e. manager's belief is correct) Perform a 2-tailed test at 5% level of significance. Test Statistics: Under H_0, $\bar{X} \sim N\left(3500, \frac{s^2}{60}\right)$ approximately $Z = \frac{\bar{X} - 3500}{s/\sqrt{60}} \sim N(0,1) \text{ approximately}$ Using GC, $p\text{-value} = 0.196284 (>0.05)$ Do not reject H_0. There is insufficient evidence at 5% level of significance that the human resource manager's belief is correct.	
	(iii) $p\text{-value}$ of 0.196284 is the smallest value of significance level for which the claim that the mean starting monthly salary of a fresh university graduate is \$3500 would be rejected. Or $p\text{-value}$ of 0.196284 is the probability of obtaining a sample mean as extreme as the one obtained, assuming the claim that the mean starting monthly salary of a fresh university graduate is \$3500 is true.	

	(iv) Test $H_0: \mu = 3500$ Against $H_1: \mu \neq 3500$ Perform a 2-tailed test at 5% level of significance. $s^2 = \frac{50}{49} \times 342^2 = 119351.0204$ Test Statistics: Under H_0, $\bar{Y} \sim N\left(3500, \frac{119351.0204}{50}\right)$ approximately $Z = \frac{\bar{Y} - 3500}{\sqrt{119024.4898/50}} \sim N(0,1) \text{ approximately}$ Since H_0 is rejected, $\frac{\bar{y} - 3500}{\sqrt{119351.0204/50}} < -1.95996$ $\bar{y} < 3404.241954$ Or $\frac{\bar{y} - 3500}{\sqrt{119351.0204/50}} > 1.95996$ $\bar{y} > 3595.758046$ Answer: $\bar{y} < 3400$ or $\bar{y} > 3600$ (to 3sf)	
	(v) $\bar{X} \sim N\left(3500, \frac{543^2}{50}\right)$ approximately by CLT since sample size (50) is large. $P(\bar{X} < 3350) = 0.0254$	

10.	Solution [13 marks]	
	(i) $\mu = \frac{60+120}{2} = 90$ minutes	

	(ii) Let C be Gary's timing for cycling. $C \sim N(90, \sigma^2)$ $Z = \frac{C - 90}{\sigma} \sim N(0, 1)$ $P(C < 80) = 0.158655$ $P\left(Z < \frac{80 - 90}{\sigma}\right) = 0.158655$ $\frac{80 - 90}{\sigma} = -1.000001057$ $\sigma = \frac{-10}{-1.000001057} \approx 10$	
	(iii) Let R be Gary's timing for running. $R \sim N(100, 15^2)$ $3C - 2R \sim N(70, 1800)$ $P(3C - 2R > 60) = 0.5931680976 \approx 0.593$ (3 sig fig)	
	(iv) Required probability $= P(C < 80) \times P(R < 90)$ $= 0.0400592579$ ≈ 0.0401 (3 sig fig)	
	(v) $C + R \sim N(190, 325)$ $P(C + R < 170) = 0.1336287896 \approx 0.134$ (3 sig fig)(shown) The event in part (iv) is but a proper subset of that in this part, hence the probability is here is greater than that in (iv).	

	(vi) Let X denote the number of duathlons which Gary manages to achieve his target timing, out of 9 duathlons. $X \sim B(9, 0.134)$ $P(3 \leq X < 6) = P(X \leq 5) - P(X \leq 2)$ $= 0.1080999944$ $\approx 0.108 \text{ (3 sig fig)}$ The timings for each of the duathlons are independent.	
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